k-Nearest Neighbors (k-NN) Classification with Iris Flower Dataset

Iris Flower Dataset

- Introduced by the British biologist and statistician Ronald A. Fisher in 1936
- Named after a flowering plant genus called Iris
- It is a well-known dataset in the field of machine learning and statistics
- Used for practicing classification and clustering algorithms
- Consists of 4 numerical features and 1 categorical feature
  - The categorical feature has 3 classes
- There are 150 samples in the dataset
  - 50 samples for each class

The three species of Iris flowers

Iris setosa

Iris versicolor

Iris virginica

Images from: https://en.wikipedia.org/wiki/Iris_flower_data_set

The four measurements for Iris flowers

These four features are the measurements used to characterize each Iris flower:

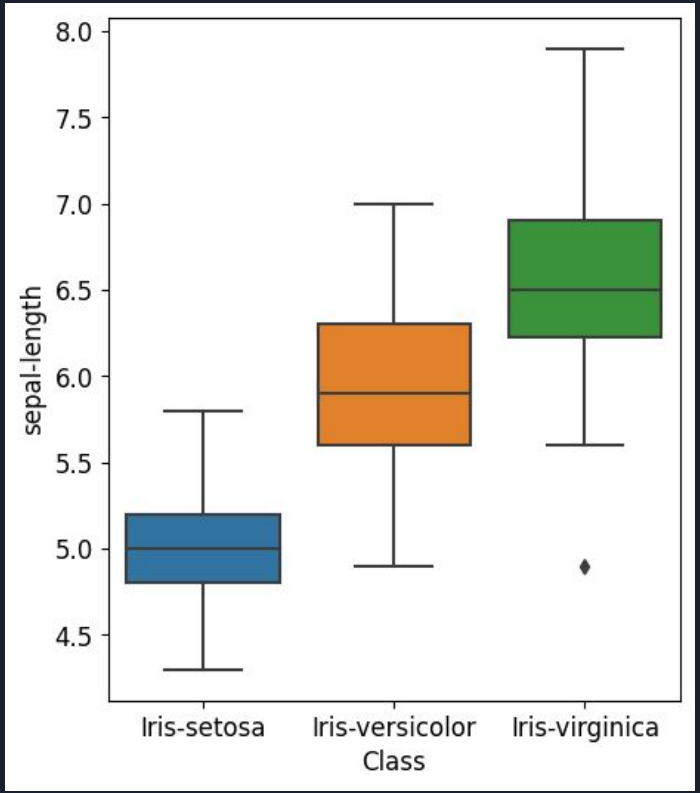
- Sepal length
- Sepal width
- Petal length
- Petal width

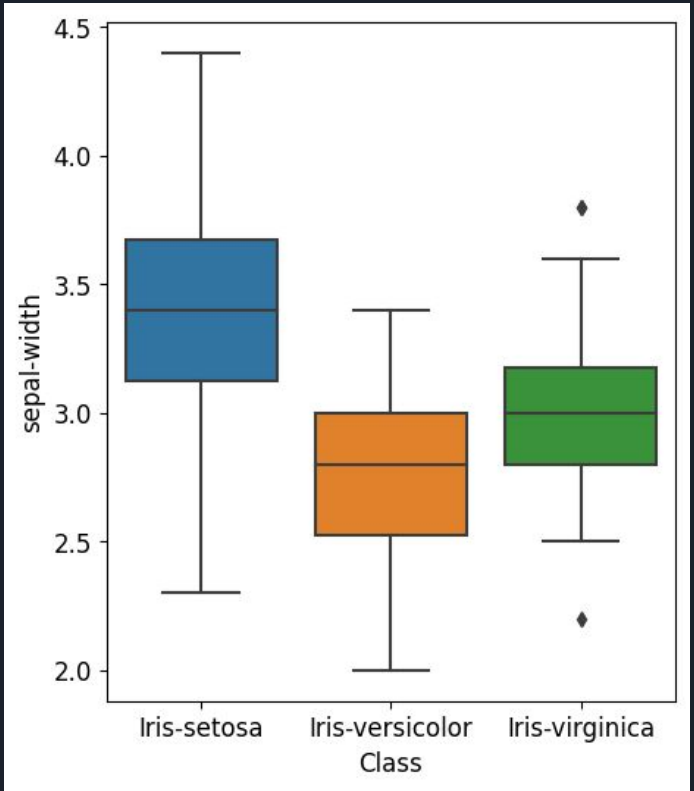
All of these are measured in centimeters

Descriptive Statistics

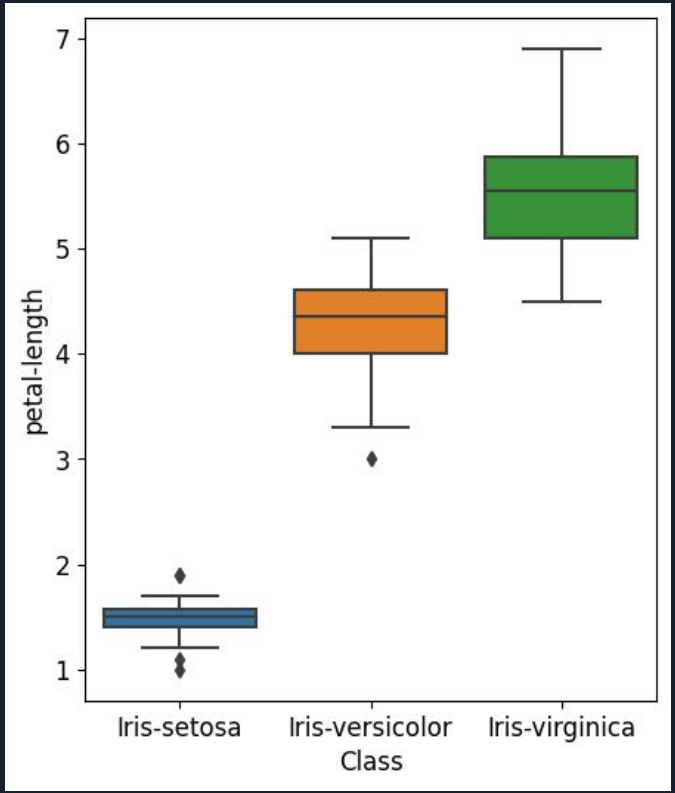
	sepal-length	sepal-width	petal-length	petal-width
mean	5.84	3.05	3.76	1.20
std	0.82	0.43	1.76	0.76
min	4.30	2.00	1.00	0.10
max	7.90	4.40	6.90	2.50

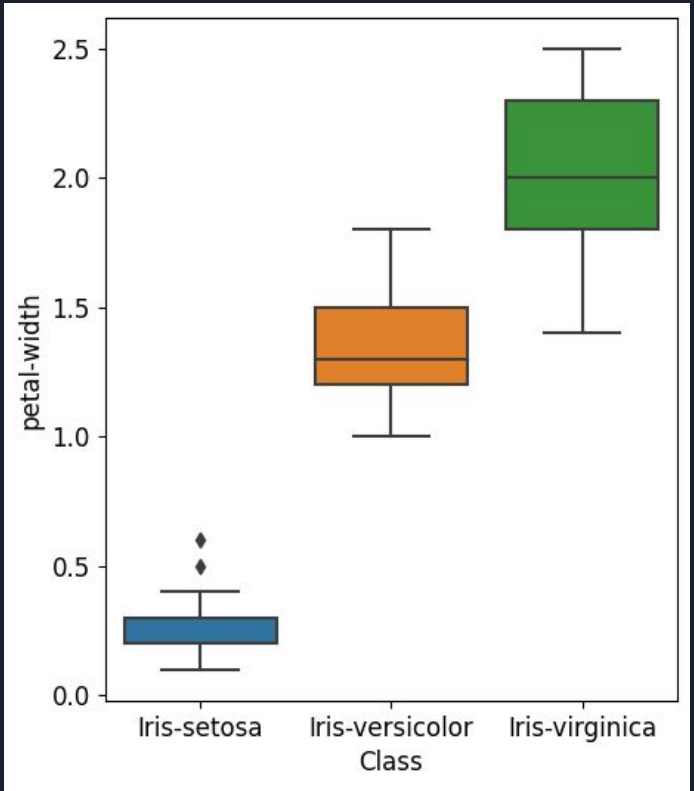
Boxplot
Sepal Length & Width



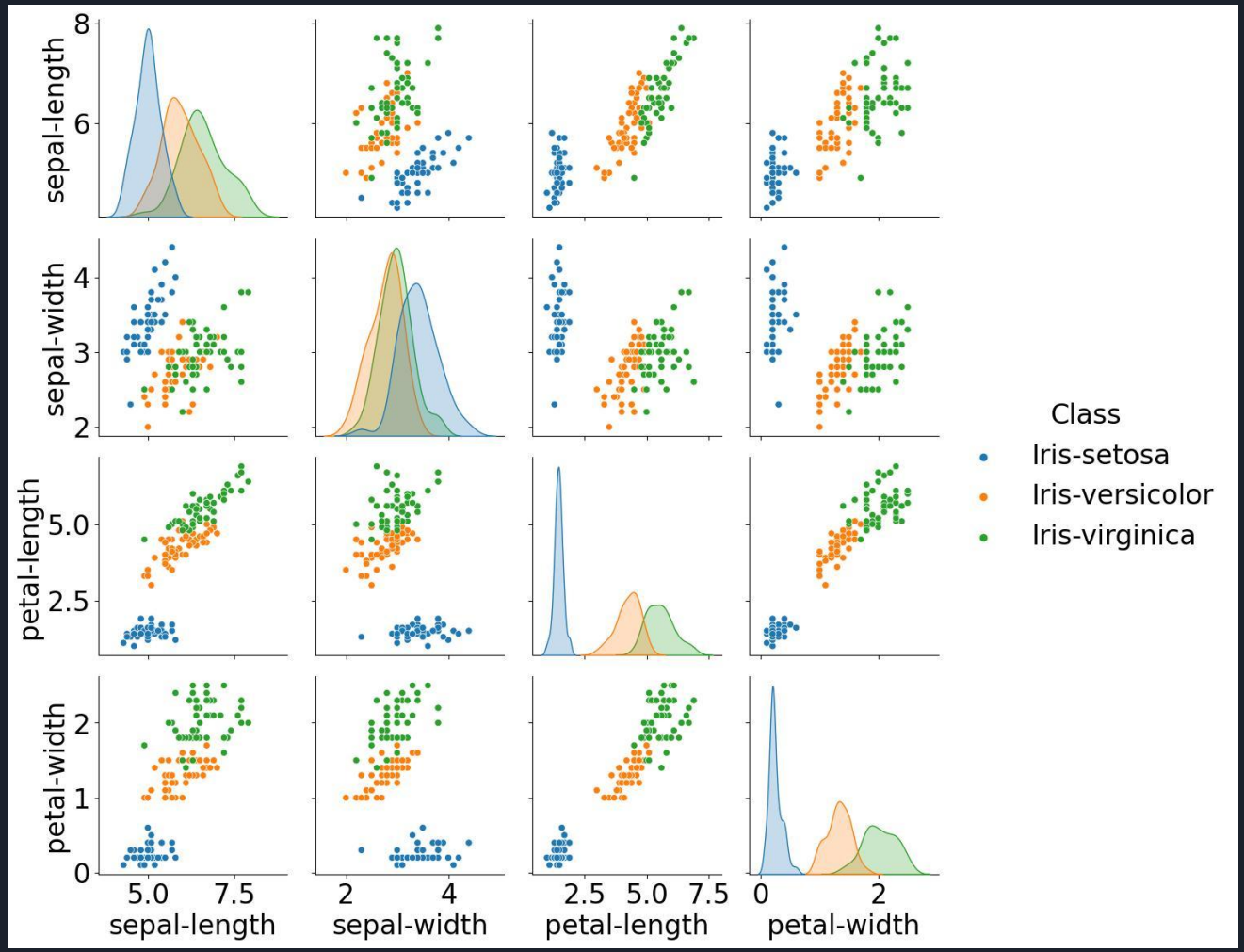


Boxplot
Petal Length & Width





Pairwise Relationship



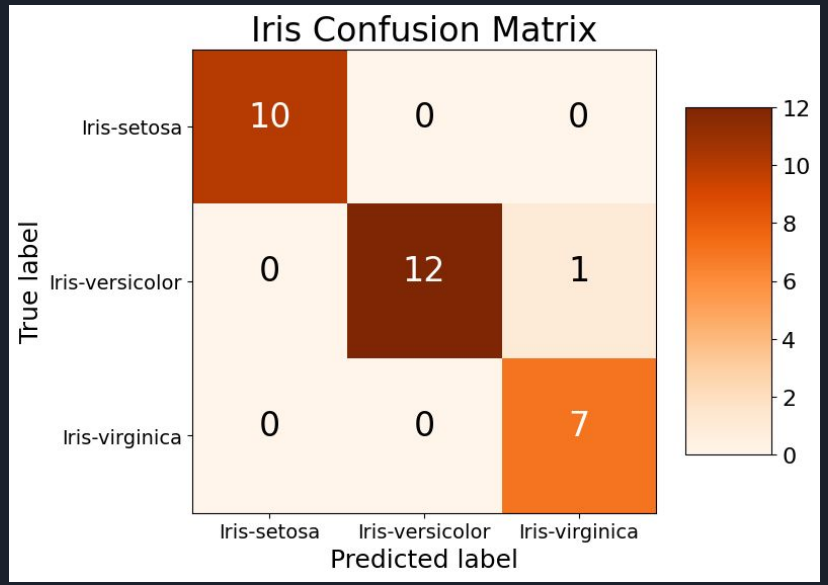
Pairwise Correlation

Classify Iris flower species using k-NN

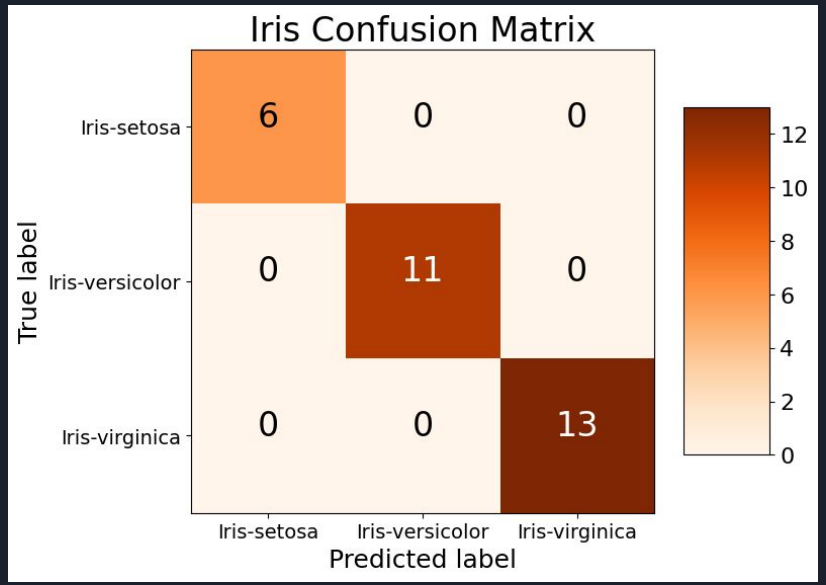
- Input (X):
  - sepal-length
  - sepal-width
  - petal-length
  - petal-width
- Target (y): classes of Iris flower
  - Iris setosa
  - Iris versicolor
  - Iris virginica
- Dataset split:
  - Train: 80%
  - Test: 20%
- k-neighbor: 5
- Run: 2

Confusion Matrices

Run 1

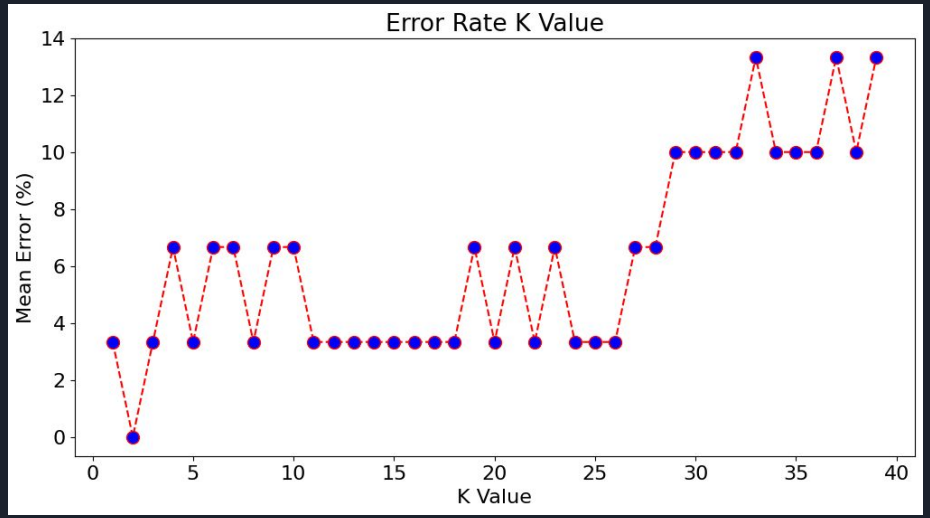


Run 2

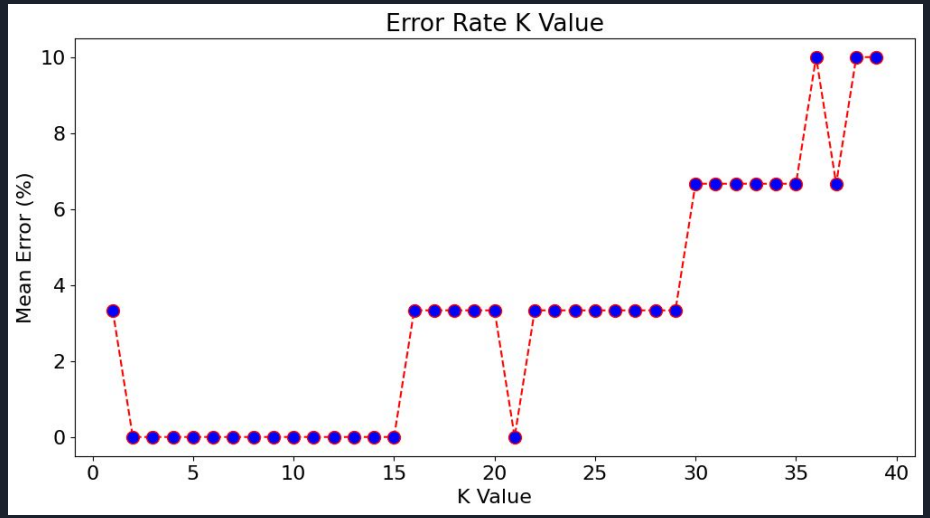


Prediction Errors

Run 1



Run 2



Why the difference? The function used to split dataset <u>shuffles</u> dataset before a split by default

Thank you for your time